

Silverline House — 25-Storey Residential HRB

Material Specification — Concrete and Steel

Document Reference: 2.1.5

Revision: Rev A

Date: March 2026

Project Reference: SLH-2026

1 Concrete Specification

All reinforced concrete elements are specified in C32/40 concrete (characteristic cylinder / cube strength) with a maximum water-cement ratio of 0.55 and minimum cement content of 300 kg/m³. Post-tensioned slab tendons are specified in low-relaxation 15.7 mm strand to BS 5896, stressed to 75% of characteristic tensile strength. Reinforcement throughout is Grade B500B to BS 4449. Concrete cover to reinforcement is 35 mm to internal elements and 50 mm to external elements exposed to the weather, in accordance with BS EN 1992-1-1 Table 4.4N for exposure class XC3. Minimum cement content and maximum water-cement ratio comply with the durability requirements of BS 8500-1 for the intended service life of 50 years. The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

2 Reinforcement Specification

All reinforcement is Grade B500B hot-rolled deformed bar to BS 4449:2005+A3. Reinforcement fabric to BS 4483. Stainless steel reinforcement to BS EN 10088-2 Grade 1.4301 is used at slab edges and in any elements exposed to de-icing salts. Post-tensioning tendons and anchorages comply with BS 4447 and BS EN 13391. Design loads have been derived in accordance with BS EN 1991-1-1 and BS EN 1991-1-3 for imposed, snow and wind loading respectively, with the UK National Annex applied throughout. The imposed load for residential floors is 1.5 kN/m² with an additional 1.0 kN/m² for partitions. Plant rooms at roof level are designed for 5.0 kN/m². Wind loading has been calculated for a basic wind speed of 23 m/s (10-minute mean at 10 m height in open country) adjusted for the site location and topography in accordance with BS EN 1991-1-4 and the UK National Annex. The net lateral wind force on the building is 3,200 kN at the 50-year return period, resisted by the reinforced concrete core.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

3 Structural Steel Specification

Structural steelwork is to S355 JR/J0/J2 to BS EN 10025-2 as appropriate for element thickness. Hollow sections are to Grade S355 J2H to BS EN 10210. Welding is in accordance with BS EN 1011-2. All structural steel connections are designed to BS EN 1993-1-8. Hot-dip galvanising to BS EN ISO 1461 is specified for all exposed external steelwork. All reinforced concrete elements are specified in C32/40 concrete (characteristic cylinder / cube strength) with a maximum water-cement ratio of 0.55 and minimum cement content of 300 kg/m³. Post-tensioned slab tendons are specified in low-relaxation 15.7 mm strand to BS 5896, stressed to 75% of characteristic tensile strength. Reinforcement throughout is Grade B500B to BS 4449. Concrete cover to reinforcement is 35 mm to internal elements and 50 mm to external elements exposed to the weather, in accordance with BS EN 1992-1-1 Table 4.4N for exposure class XC3. Minimum cement content and maximum water-cement ratio comply with the durability requirements of BS 8500-1 for the intended service life of 50 years.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.